

GRANTLEY ADAMS INTL, Barbados

WMO: 789540

Lat: 13.0853N

Lon: 59.4873W

Elev: 66

StdP: 100.53

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 00408

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.7	23.1	17.9	12.9	25.4	18.8	13.7	25.6	12.7	26.4	12.2	26.6	4.0	50	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	5.0	31.2	26.4	31.1	26.3	31.0	26.3	27.3	29.9	27.1	29.8	26.8	29.6	6.9	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.8	22.6	29.0	26.2	21.8	28.7	26.1	21.7	28.6	87.0	30.1	85.7	30.2	84.7	29.8	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.6	10.9	10.2	20.8	32.0	0.8	0.4	20.2	32.3	19.7	32.6	19.3	32.8	18.7	33.1
			18.8	28.4	0.8	0.6	18.2	28.8	17.7	29.2	17.2	29.6	16.6	30.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	26.4	26.3	26.7	27.3	28.1	28.2	28.1	28.2	28.2	28.0	27.6	27.0
	DBStd	1.08	0.78	0.83	0.89	0.88	0.76	0.73	0.71	0.84	0.97	0.90	0.86	0.80
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6387	508	457	518	521	560	544	559	563	545	557	528	527
	CDD18.3	3346	249	223	260	271	302	294	301	305	295	299	278	268
	CDH23.3	34498	2082	1812	2274	2664	3299	3268	3359	3493	3414	3353	2931	2549
(13) Wind	WSAvg	6.3	6.7	7.0	6.6	6.5	6.9	7.5	6.8	5.7	4.9	5.3	5.6	6.6
	PrecAvg	1168	67	43	38	59	53	103	125	151	149	171	156	108
	PrecMax	1854	200	104	95	305	205	240	331	336	373	457	458	279
	PrecMin	843	9	3	7	6	10	8	28	37	50	35	36	13
	PrecStd	208	39	24	20	54	48	56	65	62	77	85	95	67
	DB	30.1	30.1	30.9	31.1	31.2	31.2	31.2	31.7	32.1	31.9	31.1	30.2	(19) 30.2
	MCWB	24.6	24.7	25.1	25.8	26.0	25.9	26.1	26.5	26.8	26.7	26.6	25.4	(20) 25.4
(21) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	29.2	29.2	30.1	30.2	31.0	31.0	31.0	31.2	31.2	31.2	31.2	30.4	(21) 30.4
	MCWB	24.2	23.9	24.6	25.0	25.9	25.9	26.0	26.5	26.5	26.6	26.6	26.2	(22) 26.2
	DB	29.0	29.0	29.3	30.0	30.5	30.2	30.2	30.9	31.1	30.9	30.1	29.2	(23) 29.2
	MCWB	24.1	23.7	24.1	24.9	25.5	25.6	25.9	26.4	26.5	26.5	26.0	24.8	(24) 24.8
	DB	28.2	28.5	29.0	29.7	30.1	30.0	30.1	30.2	30.8	30.2	29.9	29.0	(25) 29.0
	MCWB	23.7	23.5	23.9	24.7	25.3	25.6	25.8	26.1	26.4	26.2	25.9	24.6	(26) 24.6
	WB	25.9	25.8	26.0	26.6	27.4	27.2	27.4	27.7	27.7	27.6	27.4	26.5	(27) 26.5
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	28.3	28.9	29.2	29.6	29.8	29.4	29.5	30.0	30.3	30.0	29.6	28.8	(28) 28.8
	WB	25.2	25.0	25.3	26.0	26.8	26.8	27.0	27.2	27.2	27.1	26.9	26.1	(29) 26.1
	MCDB	27.9	28.0	28.7	28.9	29.4	29.3	29.4	29.9	30.1	29.9	29.5	28.4	(30) 28.4
	WB	24.8	24.5	24.9	25.6	26.3	26.4	26.5	26.8	26.9	26.8	26.5	25.7	(31) 25.7
	MCDB	27.6	27.5	28.2	28.6	29.0	28.9	29.0	29.7	29.9	29.6	29.0	28.1	(32) 28.1
	WB	24.5	24.1	24.5	25.3	26.0	26.1	26.2	26.5	26.5	26.5	26.1	25.3	(33) 25.3
	MCDB	27.2	27.2	27.8	28.3	28.7	28.7	29.2	29.5	29.2	29.2	28.6	27.9	(34) 27.9
(35) Mean Daily Temperature Range	MDBR	4.7	4.9	5.0	4.8	4.4	4.3	4.4	4.8	5.0	5.0	4.7	4.6	(35) 4.6
	5% DB	4.9	5.0	5.1	5.0	4.6	4.4	4.7	5.0	5.3	5.3	4.9	4.7	(36) 4.7
	MCWBR	2.6	2.6	2.6	2.5	2.4	2.3	2.4	2.5	2.7	2.8	2.7	2.6	(37) 2.6
	5% WB	4.6	4.7	4.8	4.6	4.3	4.2	4.5	4.8	5.0	5.1	4.7	4.5	(38) 4.5
	MCWBR	2.7	2.8	2.7	2.5	2.5	2.4	2.6	2.7	2.8	2.8	2.7	2.7	(39) 2.7
(40) Clear-Sky Solar Irradiance	taub	0.376	0.393	0.390	0.434	0.477	0.561	0.519	0.459	0.422	0.385	0.379	0.385	(40) 0.385
	taud	2.575	2.497	2.524	2.406	2.263	2.002	2.135	2.367	2.488	2.625	2.625	2.572	(41) 2.572
	Ebn at Noon	926	920	927	877	826	750	786	844	883	917	917	907	(42) 907
	Edh at Noon	98	110	109	122	138	177	156	125	111	95	93	97	(43) 97
(44) All-Sky Solar Radiation	RadAvg	5.35	6.04	6.44	6.58	6.42	6.12	6.28	6.29	6.10	5.53	5.08	5.09	(44) 5.09
	RadStd	0.24	0.15	0.28	0.26	0.37	0.25	0.31	0.32	0.34	0.23	0.37	0.25	(45) 0.25

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page